

Font Stacks Documentation

This document provides a summary architectural table of font fallbacks used to ensure ideal cross-platform rendering. Fonts are grouped by operating system type, commercial vendors, and open licenses for all key `CJK` locales (China, Japan, Korea).

Font Mapping Summary Table

General Style	Region / Locale	Google Fonts (Noto)	Adobe	Apple (macOS, iOS)	Microsoft Windows	Linux / OFL
Serif/Ming	Global	Noto Serif	Source Serif	New York, Times New Roman, Times	Times New Roman	Liberation Serif, FreeSerif
	CJK JP (ja)	Noto Serif CJK JP	Source Han Serif JP	Hiragino Mincho ProN/Pro	MS PMincho, MS Mincho	IPAexMincho, IPAMincho
	CJK SC (zh-CN)	Noto Serif CJK SC	Source Han Serif SC/CN	Songti SC, STSong	SimSun	FandolSong, WenQuanYi Bitmap
	CJK TC (zh-TW)	Noto Serif CJK TC	Source Han Serif TC/TW	Apple LiSung, LiSong Pro	PMingLiU, MingLiU	HanaMinA (花園明朝)
	CJK HK (zh-HK)	Noto Serif CJK HK	Source Han Serif HK	Apple LiSung, LiSong Pro	MingLiU_HKSCS, PMingLiU	HanaMinA (花園明朝)
	CJK KR (ko)	Noto Serif CJK KR	Source Han Serif KR	AppleMyungjo	Batang	UnBatang (은바탕)
Sans-Serif/Gothic	Global	Noto Sans	Source Sans	SF Pro, Helvetica, Arial	Segoe UI, Arial	Liberation Sans, Arima, Open Sans
	CJK JP (ja)	Noto Sans CJK JP	Source Han Sans JP	Hiragino Kaku Gothic ProN	Meiryo, Yu Gothic	IPAexGothic, IPAGothic
	CJK SC (zh-CN)	Noto Sans CJK SC	Source Han Sans SC/CN	PingFang SC, STHeiti	Microsoft YaHei	FandolHei, WenQuanYi Zen Hei
	CJK TC (zh-TW)	Noto Sans CJK TC	Source Han Sans TC/TW	PingFang TC, LiHei Pro	Microsoft JhengHei	Uming (文鼎自由字型)
	CJK HK (zh-HK)	Noto Sans CJK HK	Source Han Sans HK	PingFang HK	Microsoft JhengHei UI	Uming (文鼎自由字型)
	CJK KR (ko)	Noto Sans CJK KR	Source Han Sans KR	Apple SD Gothic Neo	Malgun Gothic	UnDotum (은돌움)
Monospace/Mono	Global	Noto Sans Mono	Source Code Pro	SF Mono, Courier New, Courier	Cascadia Code, Courier New, Consolas	Liberation Mono, Fira Code
	CJK JP (ja)	Noto Sans Mono CJK JP	Source Han Mono JP	Osaka-Mono, Hiragino Mono	MS Gothic, Yu Gothic UI	IPAGothic (Mono)

General Style	Region / Locale	Google Fonts (Noto)	Adobe	Apple (macOS, iOS)	Microsoft Windows	Linux / OFL
	CJK SC (zh-CN)	Noto Sans Mono CJK SC	Source Han Mono SC/CN	PingFang SC (Mono)	NSimSun, SimHei	FandolMono, Sazanami Mono
	CJK TC (zh-TW)	Noto Sans Mono CJK TC	Source Han Mono TC/TW	PingFang TC (Mono)	MingLiU	HanaMinA (Mono)
	CJK HK (zh-HK)	Noto Sans Mono CJK HK	Source Han Mono HK	PingFang HK (Mono)	MingLiU_HKSCS	HanaMinA (Mono)
	CJK KR (ko)	Noto Sans Mono CJK KR	Source Han Mono KR	Apple SD Gothic Neo (Mono)	GulimChe, DotumChe	UnDotum (Mono)
Emoji	Global	Noto Color Emoji	EmojiOne Color, Source Emoji	Apple Color Emoji	Segoe UI Emoji, Segoe UI Symbol	Symbola

Architectural Principles of Building Stacks

- Latin → Regional Hierarchy:** Global Latin fonts are always listed first in the CSS rule (`font-family: Global, CJK, ...`). Since Latin fonts do not contain Asian ideographs, the browser seamlessly hands off the rendering of CJK glyphs to the regional fonts next in the chain.
- Style Group Purity:** The **Serif/Ming** styles include exclusively print-style serif fonts, preventing an accidental fallback to the calligraphic (handwritten) Kaiti style on older Windows versions.
- Proportional vs. Monospaced Split (Windows):**
 - Proportional-width fonts (containing the letter **"P"**, e.g. `PMingLiU` , `MS PMincho`) are chosen for body text.
 - Strictly monospaced versions without the letter **"P"**, or fonts ending in **"Che"** (`GulimChe`), are used for code.

Optimized CSS Rules

- The `@import` directive is used to load **Google Fonts** automatically right inside your CSS file (without the need to edit the page's HTML). Add this block at the very beginning of the main CSS file.
- Important:** per the CSS specification, `@import` directives must be placed *strictly before any other style rules* (including variables and selectors).

```
/* =====  
NOTO FONTS FROM GOOGLE FONTS  
===== */  
  
/* 1. Global Latin Noto fonts (Serif, Sans, Mono) */  
@import url('https://googleapis.com');  
  
/* 2. Asian Noto Serif CJK packages (Japan, China, Taiwan, Hong Kong, Korea) */  
@import url('https://googleapis.com');  
  
/* 3. Asian Noto Sans CJK packages (Japan, China, Taiwan, Hong Kong, Korea) */  
@import url('https://googleapis.com');  
  
/* 4. Asian monospaced Noto Sans Mono CJK packages */  
@import url('https://googleapis.com');  
  
/* =====  
YOUR OPTIMIZED CSS RULES GO BELOW...  
===== */
```

Optimized CSS rules:

```

/* =====
1. SERIF/MING GROUP (Print serif fonts)
===== */

/* Global */
.font-serif {
    font-family: "Noto Serif", "Source Serif", "New York", "Times New Roman", Times, "Liberation Serif", "FreeSerif", -apple-system-serif, serif;
}
/* CJK JP (ja) */
.font-serif [lang="ja"] {
    font-family: "Noto Serif CJK JP", "Source Han Serif JP", "Hiragino Mincho ProN", "Hiragino Mincho Pro", "MS PMincho", "MS Mincho", "IPAexMincho",
    "IPAMincho", serif;
}
/* CJK SC (zh-CN) */
.font-serif [lang="zh-CN"], .font-serif [lang="zh-Hans"] {
    font-family: "Noto Serif CJK SC", "Source Han Serif SC", "Source Han Serif CN", "Songti SC", "STSong", "SimSun", "FandolSong", "WenQuanYi Bitmap", serif;
}
/* CJK TC (zh-TW) */
.font-serif [lang="zh-TW"], .font-serif [lang="zh-Hant"] {
    font-family: "Noto Serif CJK TC", "Source Han Serif TC", "Source Han Serif TW", "Apple LiSung", "LiSong Pro", "PMingLiU", "MingLiU", "HanaMinA", serif;
}
/* CJK HK (zh-HK) */
.font-serif [lang="zh-HK"] {
    font-family: "Noto Serif CJK HK", "Source Han Serif HK", "Apple LiSung", "LiSong Pro", "MingLiU_HKSCS", "PMingLiU", "MingLiU", "HanaMinA", serif;
}
/* CJK KR (ko) */
.font-serif [lang="ko"] {
    font-family: "Noto Serif CJK KR", "Source Han Serif KR", "AppleMyungjo", "Batang", "UnBatang", serif;
}

/* =====
2. SANS-SERIF/GOTHIC GROUP (Sans-serif fonts)
===== */

/* Global */
.font-sans {
    font-family: "Noto Sans", "Source Sans", "SF Pro", "Segoe UI", "Open Sans", "Helvetica", Arial, "Liberation Sans", "Arima", sans-serif;
}
/* CJK JP (ja) */

```

```

.font-sans [lang="ja"] {
  font-family: "Noto Sans CJK JP", "Source Han Sans JP", "Hiragino Kaku Gothic ProN", "Meiryo", "Yu Gothic", "IPAexGothic", "IPAGothic", sans-serif;
}
/* CJK SC (zh-CN) */
.font-sans [lang="zh-CN"], .font-sans [lang="zh-Hans"] {
  font-family: "Noto Sans CJK SC", "Source Han Sans SC", "Source Han Sans CN", "PingFang SC", "STHeiti", "Microsoft YaHei", "FandolHei", "WenQuanYi Zen
Hei", sans-serif;
}
/* CJK TC (zh-TW) */
.font-sans [lang="zh-TW"], .font-sans [lang="zh-Hant"] {
  font-family: "Noto Sans CJK TC", "Source Han Sans TC", "Source Han Sans TW", "PingFang TC", "LiHei Pro", "Microsoft JhengHei", "Uming", sans-serif;
}
/* CJK HK (zh-HK) */
.font-sans [lang="zh-HK"] {
  font-family: "Noto Sans CJK HK", "Source Han Sans HK", "PingFang HK", "Microsoft JhengHei UI", "Uming", sans-serif;
}
/* CJK KR (ko) */
.font-sans [lang="ko"] {
  font-family: "Noto Sans CJK KR", "Source Han Sans KR", "Apple SD Gothic Neo", "Malgun Gothic", "UnDotum", sans-serif;
}

/* =====
3. MONOSPACE/MONO GROUP (Monospaced fonts for code)
===== */

/* Global */
.font-mono {
  font-family: "Noto Sans Mono", "Source Code Pro", "SF Mono", "Cascadia Code", "Fira Code", "Courier New", Courier, "Consolas", "Liberation Mono",
monospace;
}
/* CJK JP (ja) */
.font-mono [lang="ja"] {
  font-family: "Noto Sans Mono CJK JP", "Source Han Mono JP", "Osaka-Mono", "Hiragino Mono", "MS Gothic", "Yu Gothic UI", "IPAGothic", monospace;
}
/* CJK SC (zh-CN) */
.font-mono [lang="zh-CN"], .font-mono [lang="zh-Hans"] {
  font-family: "Noto Sans Mono CJK SC", "Source Han Mono SC", "Source Han Mono CN", "PingFang SC", "NSimSun", "SimHei", "FandolMono", "Sazanami Mono",
monospace;
}
/* CJK TC (zh-TW) */

```

```

.font-mono [lang="zh-TW"], .font-mono [lang="zh-Hant"] {
  font-family: "Noto Sans Mono CJK TC", "Source Han Mono TC", "Source Han Mono TW", "PingFang TC", "MingLiU", "HanaMinA", monospace;
}
/* CJK HK (zh-HK) */
.font-mono [lang="zh-HK"] {
  font-family: "Noto Sans Mono CJK HK", "Source Han Mono HK", "PingFang HK", "MingLiU_HKSCS", "HanaMinA", monospace;
}
/* CJK KR (ko) */
.font-mono [lang="ko"] {
  font-family: "Noto Sans Mono CJK KR", "Source Han Mono KR", "Apple SD Gothic Neo", "GulimChe", "DotumChe", "UnDotum", monospace;
}

/* =====
4. EMOJI GROUP (Standalone emojis and pictograms)
===== */

.emoji {
  font-family: "Noto Color Emoji", "Apple Color Emoji", "Segoe UI Emoji", "EmojiOne Color", "Source Emoji", "Segoe UI Symbol", "Symbola", sans-serif;
  font-style: normal;
}

```

Principles for Creating CSS Rules

- **Noto** family fonts (both global and regional CJK) are moved to the front of every CSS rule (the browser will be forced to look up and use **Google's** fonts).
- The browser will fall back to system alternatives from **Apple (macOS)** or **Microsoft (Windows)** only if the **Noto** web fonts are not loaded via the `<link>` tag, did not download, or are not installed locally.

The table has been turned into clean CSS rules. They automatically pick up the text language via the standard `[lang="..."]` attribute.

Within each rule, fonts are ordered by the following priority:

1. Exact **localized web font names** (e.g. "Noto Serif CJK JP").
2. Region-specific **Apple** system fonts.
3. Region-specific **MS Windows** system fonts (proportional first, then monospaced).
4. Free **Linux / OFL** fonts.
5. Generic family (serif , sans-serif , monospace).